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Scope(s): Global
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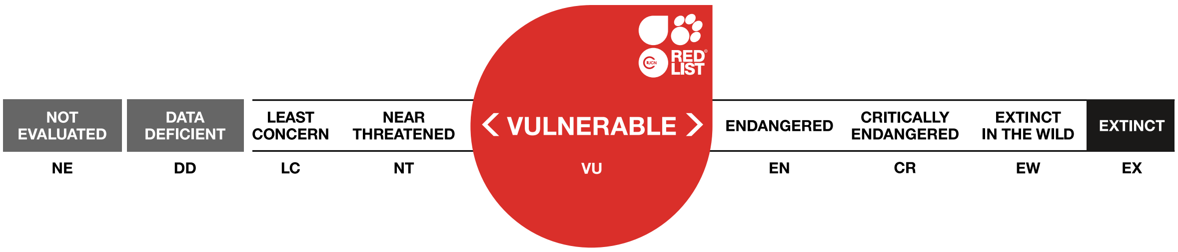
Carcharodon carcharias, White Shark

Amendment version

Assessment by: Rigby, C.L. *et al.*



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Chondrichthyes	Lamniformes	Lamnidae

Scientific Name: *Carcharodon carcharias* (Linnaeus, 1758)

Synonym(s):

- *Squalus carcharias* Linnaeus, 1758

Regional Assessments:

- Mediterranean
- Europe

Common Name(s):

- English: White Shark, Great White Shark

Taxonomic Source(s):

Fricke, R., Eschmeyer, W.N. and Van der Laan, R. (eds). 2019. Eschmeyer's Catalog of Fishes: genera, species, references. Updated 03 September 2019. Available at: http://researcharchive.calacademy.org/research/ichthyology/catalog/fishcatmain.asp.

Assessment Information

Red List Category & Criteria: Vulnerable A2bd ver 3.1

Year Published: 2022

Date Assessed: November 7, 2018

Justification:

The White Shark (*Carcharodon carcharias*) is a large (to 640 cm total length) coastal and pelagic shark, wide-ranging throughout most temperate and tropical oceans to depths of 1,200 m. The species is very long-lived, up to 73 years, with females not reaching maturity until 33 years, resulting in a long generation length of 53 years. It is caught globally, mostly as bycatch in inshore fisheries by a range of gears, is rarely caught in offshore pelagic fisheries, and is targeted in beach protection programs in Australia and South Africa; however, in some instances these programs release live sharks. Over three generation lengths (159 years), the White Shark is estimated to be increasing in abundance in the Northeast Pacific and Indian Ocean, and declining in abundance from historic levels in the Northwest Atlantic and South Pacific. In most regions, declines occurred during the 1980s followed by slow recovery since the 1990s when protection was implemented. Globally, based on long-term abundance data and protections instigated in the 1990s that have since reduced catches and allowed some recovery, the White Shark population is estimated to have reduced by 30–49% over the last three generations (159 years), and is therefore assessed as Vulnerable A2bd.

Previously Published Red List Assessments

2019 – Vulnerable (VU)

https://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T3855A2878674.en

2009 – Vulnerable (VU)
https://dx.doi.org/10.2305/IUCN.UK.2009-2.RLTS.T3855A10133872.en

2000 – Vulnerable (VU)

1996 – Vulnerable (VU)

1994 – Insufficiently Known (K)

1990 – Insufficiently Known (K)

Geographic Range

Range Description:

The White Shark is wide-ranging throughout most temperate and tropical oceans but occurs most frequently in temperate waters (Last and Stevens 2009, Ebert *et al.* 2013).

Country Occurrence:

Native, Extant (resident): Albania; Algeria; American Samoa; Angola; Anguilla; Argentina; Australia; Bahamas; Benin; Bermuda; Bonaire, Sint Eustatius and Saba (Saba, Sint Eustatius); Bosnia and Herzegovina; Brazil; Canada; Chile; China; Congo; Cook Islands; Croatia; Cuba; Cyprus; Côte d'Ivoire; Ecuador (Ecuador (mainland), Galápagos); Fiji; France (France (mainland)); French Polynesia; Gambia; Ghana; Gibraltar; Greece; Guernsey; Guinea; Guinea-Bissau; Hong Kong; Iran, Islamic Republic of; Iraq; Italy; Jamaica; Japan; Jersey; Kenya; Kiribati; Korea, Democratic People's Republic of; Korea, Republic of; Lebanon; Liberia; Libya; Macao; Madagascar; Malta; Mauritania; Mexico; Monaco; Montenegro; Morocco; Mozambique; Namibia; New Caledonia; New Zealand; Nigeria; Norfolk Island; Panama; Peru; Philippines; Pitcairn; Portugal (Azores, Madeira, Portugal (mainland)); Puerto Rico (Puerto Rico (main island)); Russian Federation; Saint Kitts and Nevis; Saint Martin (French part); Saint Pierre and Miquelon; Saint Vincent and the Grenadines; Samoa; Saudi Arabia; Senegal; Seychelles; Sierra Leone; Sint Maarten (Dutch part); Slovenia; Somalia; South Africa; Spain (Spain (mainland)); Sri Lanka; Syrian Arab Republic; Tanzania, United Republic of; Togo; Tokelau; Tunisia; Turkey; United States (Hawaiian Is.); Uruguay; Vanuatu; Viet Nam; Virgin Islands, British; Virgin Islands, U.S.; Western Sahara

Native, Possibly Extant (resident): Antigua and Barbuda; Aruba; Bahrain; Bangladesh; Barbados; Belize; Bonaire, Sint Eustatius and Saba (Bonaire); British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; Cabo Verde; Cambodia; Cameroon; Cayman Islands; Christmas Island; Cocos (Keeling) Islands; Colombia (Colombia (mainland), Colombian Caribbean Is.); Comoros; Costa Rica; Curaçao; Disputed Territory (Paracel Is., Spratly Is.); Djibouti; Dominica; Dominican Republic; El Salvador; Equatorial Guinea (Annobón, Equatorial Guinea (mainland)); Eritrea; Falkland Islands (Malvinas); France (Clipperton I.); French Guiana; Gabon; Grenada; Guadeloupe; Guam; Guatemala; Guyana; Haiti; Honduras; India; Indonesia; Malaysia; Maldives; Marshall Islands; Martinique; Mauritius; Mayotte; Micronesia, Federated States of ; Montserrat; Myanmar; Nauru; Nicaragua; Niue; Northern Mariana Islands; Oman; Pakistan; Palau; Papua New Guinea; Portugal (Selvagens); Puerto Rico (Navassa I.); Qatar; Réunion; Saint Barthélemy; Saint Helena, Ascension and Tristan da Cunha (Ascension, Saint Helena (main island), Tristan da Cunha); Saint Lucia; Sao Tome and Principe; Singapore; Solomon Islands; Spain (Canary Is.); Sudan; Suriname; Thailand; Timor-Leste; Tonga; Trinidad and Tobago; Turks and Caicos Islands; Tuvalu; United Arab Emirates; United States Minor Outlying Islands (Howland-Baker Is., Johnston I., Midway Is., US Line Is., Wake Is.); Venezuela, Bolivarian Republic of; Wallis and Futuna; Yemen

FAO Marine Fishing Areas:

Native: Atlantic - northeast

Native: Atlantic - southeast

Native: Atlantic - western central

Native: Atlantic - northwest

Native: Atlantic - eastern central

Native: Pacific - western central

Native: Pacific - southwest

Native: Pacific - southeast

Native: Pacific - northwest

Native: Pacific - northeast

Native: Atlantic - southwest

Native: Pacific - eastern central

Native: Mediterranean and Black Sea

Native: Indian Ocean - western

Native: Indian Ocean - eastern

Native: Atlantic - northeast

Native: Atlantic - southeast

Native: Atlantic - western central

Native: Atlantic - northwest

Native: Atlantic - eastern central

Native: Pacific - western central

Native: Pacific - southwest

Native: Pacific - southeast

Native: Pacific - northwest

Native: Pacific - northeast

Native: Atlantic - southwest

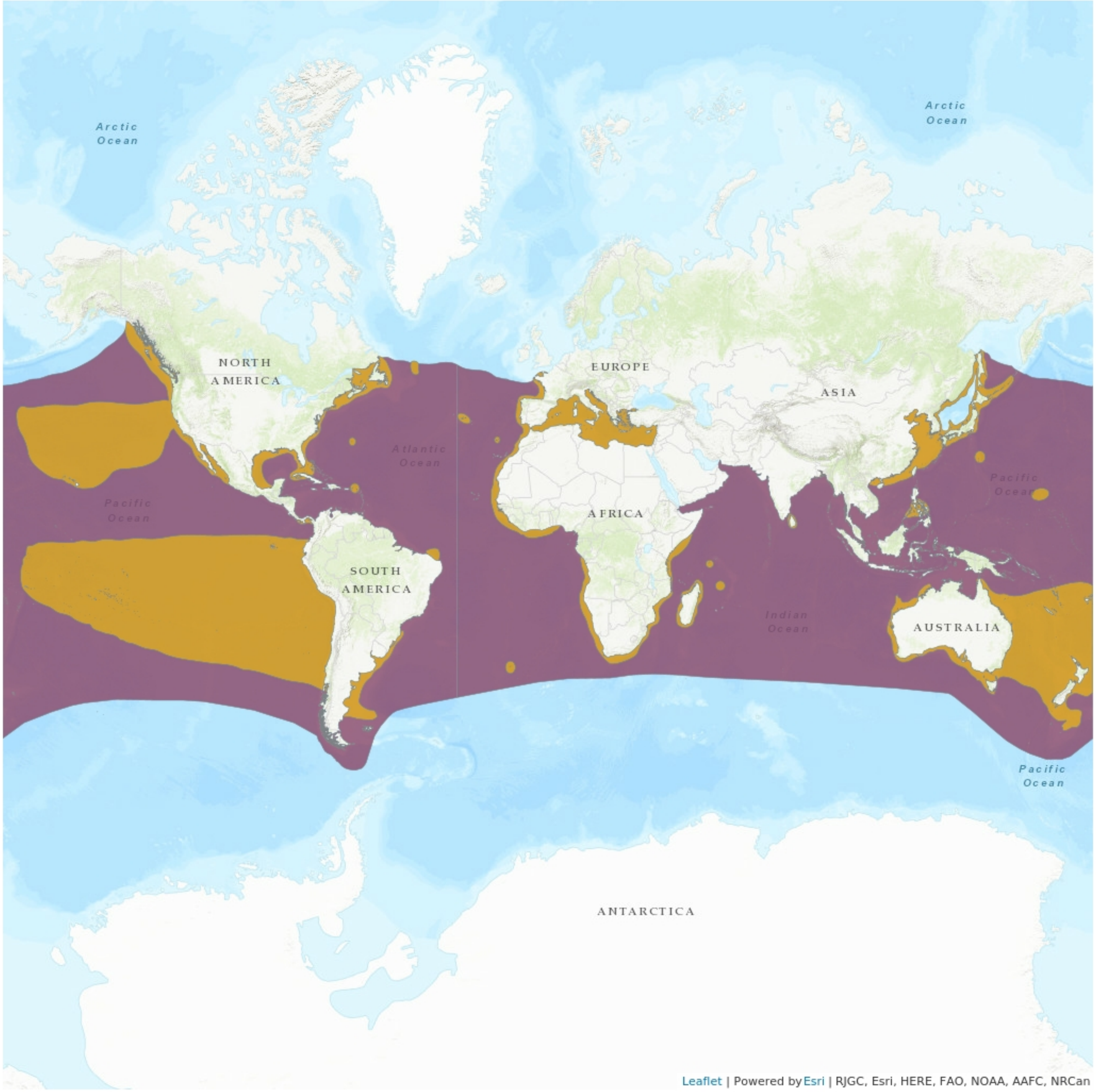
Native: Pacific - eastern central

Native: Mediterranean and Black Sea

Native: Indian Ocean - western

Native: Indian Ocean - eastern

Distribution Map



Legend

EXTANT (RESIDENT)

POSSIBLY EXTANT (RESIDENT)

Compiled by:

IUCN SSC Shark Specialist Group 2018

The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

There are no data available on the absolute global population size of the White Shark. Genetic data suggest one global population; however, there is some genetic structuring between ocean basins, potentially within ocean basins, and likely global male-biased dispersal and female philopatry (Pardini *et al*. 2001, Jorgensen *et al.* 2010, Gubili *et al*. 2012, O'Leary *et al*. 2015, Andreotti *et al.* 2016, Bernand *et al*. 2018). White Shark total abundances have been estimated for a number of regions: 5,460 in eastern Australasia in 2017 (uncertainty range: 2,909–12,802) (Bruce *et al.* 2018, Hillary *et al.* 2018); in excess of 2,000 in the Northeast Pacific in 2012 (Dewar *et al.* 2013, Burgess *et al*. 2014); and, in South Africa: 1,279 in KwaZulu-Natal in 1996 (Cliff *et al.* 1996), 908 in Gansbaai from 2007–2011 (95% CI: 808–1,008) (Towner *et al*. 2013), 723 in False Bay from 2004–2012 (95% CI: 466–980) (Hewitt 2014), 389 from 2008–2010 in Mossel Bay (95% CI: 351–428) (Ryklief 2012), and 438 for all of South Africa in 2011 (Andreotti *et al*. 2016), although there are concerns for the validity of this last estimate (Irion *et al.* 2017).

Population trend data are available from four sources: (1) standardized relative abundance in the Northwest Atlantic (Curtis *et al.* 2014); and standardized catch-per-unit-effort (CPUE) in: (2) the Northeast Pacific (Dewar *et al.* 2013); (3) the South Pacific (Reid *et al*. 2011); and, (4) the Indian Ocean (Dudley and Simpfendorfer 2006) updated with data to 2012 (S. Wintner pers. comm. 3/10/2018). The trend data from each source were analysed over three generation lengths using a Bayesian state-space framework (a modification of Winker *et al*. 2018). This analysis yields an annual rate of change, a median change over three generation lengths, and the probability of the most likely IUCN Red List Category percent change over three generations (see the Supplementary Information). First, the Curtis *et al.* (2014) relative abundance was used to represent data on catches in the Northwest Atlantic as it is the longest and most comprehensive compilation of data available for the region. It includes fishery-independent longline surveys, observer data from the shark-target bottom longline fishery, and recreational fishing tournament data. The relative abundance trend indicates historically higher abundances in the 1960s followed by a declining trend until the mid-1980s, and then an increasing trend since the 1990s, when a range of management measures were implemented (Curtis *et al.* 2014). The three generation trend analysis of the Northwest Atlantic CPUE for 1961–2010 (50 years) revealed annual rates of reduction of 1.0%, consistent with an estimated median reduction of 80.8% over three generation lengths (159 years), with the highest probability of >80% reduction over three generation lengths. This probability of a high level of reduction is over a very long period of three generations; it has incorporated the reductions from historically higher abundances and projected an estimated trend based on those reductions for a considerable period beyond that of the time-series and is thus indicative of historic declines rather than the recent increasing trend since 1990.

Second, in the Northeast Pacific, White Sharks have mainly been taken in inshore net fisheries. Fisheries catch data were available from the California set net fishery which accounts for the majority of White Shark bycatch in that area; this indicated declines during the 1980s followed by a gradual increase from 1990s onwards (Dewar *et al.* 2013). Based on those data, information from photo-identification studies, and other researchers, the status review of White Shark in the Northeast Pacific concluded that the White Shark abundance was stable or increasing although there was some uncertainty around this conclusion due to the limited long-term abundance data for the region (Dewar *et al.* 2013). The trend analysis of the Northeast Pacific California set net fishery CPUE for 1980–2010 (31 years) revealed annual rates of increase of 4.1%, consistent with an estimated median increase of 602.1% over three generation lengths (159 years), with the highest probability of increases over three generation lengths.

Third, one of the longest datasets in the South Pacific is from the east coast of Australia shark meshing program in New South Wales (1950–2009) in which White Shark standardized CPUE markedly declined from 1950 to the 1990s after which it began to slowly increase (Reid *et al. 2*011). During this period there were modifications to net specifications and spatial and temporal effort, and since 2010, further changes have been implemented that include a reduction in net soak time (Reid *et al*. 2011, NSW DPI 2018). From 2008 to 2018, catches of White Sharks in the meshing program have fluctuated between 3 and 26 individuals annually, with increasing catches in the latter years (NSW DPI 2018). Further north along Australia's east coast, the standardized CPUE of White Sharks also declined significantly, by 92% over 54 years (1962–2015) in the Queensland Shark Control Program (Roff *et al.* 2018). The trend analysis of the NSW shark meshing program CPUE for 1950–2009 (60 years) revealed annual rates of reduction of 1.8%, consistent with an estimated median reduction of 95.8% over three generation lengths (159 years), with the highest probability of >80% reduction over three generation lengths. Since mid-1990s protection of White Sharks in Australia, adult White Sharks abundance has been estimated to have slightly declined or remained stable with population growth rates unlikely to be greater than 3% per year (Bruce *et al.* 2018, Hillary *et al.* 2018). Based on the current eastern Australasian abundance estimate and species demographics (assuming continued protection) there should potentially be an increase in the eastern Australasian abundance as the current juvenile cohorts enter into maturity (R. Bradford pers. comm. 07/02/2019).

Fourth, long-term standardized CPUE in shark nets off KwaZulu-Natal, South Africa beaches in the Western Indian Ocean fluctuated considerably but was stable over time (Dudley and Simpfendorfer 2006). The trend analysis of the shark net CPUE for 1978–2012 (35 years) revealed annual rates of increase of 0.1%, consistent with an estimated median increase of 13.1% over three generation lengths (159 years), with the highest probability of increases over three generation lengths. In the West Australian Indian Ocean, the White Shark modeled abundance is not predicted to have increased by more than 10% since 1997 when protection was enacted (Braccini *et al.* 2017).

Further to the above data and analyses, in the Mediterranean Sea, based on anecdotal records and limited fisheries data, the White Shark is suspected to have declined by at least 80% over 69 years from 1947–2016 (Soldo *et al.* 2016). With the exception of the Mediterranean Sea, in all other regions with data, the status review and studies indicate that the White Shark has declined during the 1980s and has begun to show signs of increasing since protection has been implemented.

Across the regions, the White Shark is estimated to be declining from historic levels in the Northwest Atlantic and South Pacific, and increasing in the Northeast Pacific and Indian Ocean. The trends among ocean regions are highly variable and while they are mostly based on long datasets, they are extrapolated over a very long three generation length of 159 years which increases the uncertainty in the estimated regional trends. With the exception of the Northwest Atlantic, they are also based on datasets from limited areas within each region and may not accurately represent the trend in White Shark across the entire region. Despite these caveats, the trend data are the best available and were used for the estimation of a global population trend; the estimated three generation population trend for each region was weighted according to the relative size of each region. The overall estimated median reduction was 53.8%, with the highest probability of a <20% reduction over three generation lengths (159 years). With a shorter three generation length of 88.5 years (see Habitat and Ecology section), the overall estimated median reduction was 39.4%, with the highest probability of a <20% reduction over three generation lengths. However, due to uncertainty in the estimated trends, expert judgement elicitation resulted in an estimated global population reduction of 30–49% over the last three generations (159 years), based on long-term abundance data and protections instigated in the 1990s that have since reduced catches. Therefore, the White Shark is assessed as Vulnerable A2bd.

For further information about this species, see Supplementary Material.

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

The White Shark is pelagic and most commonly occurs in temperate continental shelf waters but also ranges into estuaries and the open ocean, and occurs to depths of 1,200 m (Francis *et al.* 2012). Movement studies and DNA analyses have demonstrated that the White Shark undertakes long distance trans-oceanic movements, for example between South Africa and Australasia (Pardini *et al.* 2001, Bonfil *et al.* 2005), and California and the Hawaiian Islands (Boustany *et al.* 2002, Domeier and Nasby-Lucas 2008, Jorgensen *et al*. 2010). Consequently, its distribution is not considered disjunct, rather that interchange between some areas may be limited.

The maximum size is undetermined, but estimated at 600–640 cm total length (TL) (Compagno 2001). Males mature at 310–410 cm TL (Pratt 1996, Compagno 2001, Tanaka *et al.* 2011); the majority of females mature at 400–500 cm TL (Francis 1996, Tanaka *et al.* 2011) but have been reported as immature at 472–490 cm TL (Springer 1939, Compagno 2001); and, size at birth is 120–150 cm TL (Francis 1996). Reproduction is aplacental viviparous with oophagy and histrophy, with litter sizes of 2–17 and a suspected two to three year reproductive cycle (Francis 1996, Uchida *et al.* 1996, Mollet and Cailliet 2002, Bruce 2008, Domeier 2012, Sato *et al*. 2016). In the Northwest Atlantic, Northeast Pacific, and Western Indian Oceans, female age-at-maturity of 30–33 years and maximum age of 30–73 years were reported based on bomb radiocarbon validated ages (Hamady *et al.* 2014, Andrews and Kerr 2015, Natanson and Skomal 2015, Christiansen *et al.* 2016). Due to the accurate method of bomb radiocarbon ageing, these are possibly more accurate than the previous lower estimates of female age-at-maturity of 7–15 years and maximum ages of 18–30 years (Cailliet *et al.* 1985, Wintner and Cliff 1999, Bruce 2008, Tanaka *et al.* 2011, Hillary *et al.* 2018). Using the precautionary approach and the validated, most conservative bomb radiocarbon ages of age-at-maturity of 33 years and maximum age of 73 years, generation length is 53 years. The bomb radiocarbon maximum age is accepted as being likely >30 years, minimum of 44 years, and possibly maximum of 73 years. The female age-at-maturity is possibly more contentious and to explore the sensitivity of the global population trend analyses to the younger female age-at-maturity, estimates of female age-at-maturity of 15 years and maximum age of 44 years were also used, that led to a generation length of 29.5 years (see Population section).

Systems: Marine

Use and Trade (see Appendix for additional information)

The White Shark fins and jaws have a high market value with large fins used as display items (Clarke

2004). Small white shark fins are also present in the international fin trade (Shivji *et al.* 2005). Jaws may be retained domestically as curios (CITES 2004). The meat may be used fresh for either local consumption or exported internationally.

Threats (see Appendix for additional information)

The White Shark is caught as bycatch mostly in inshore fisheries in a range of gears, such as longlines, setlines, gillnets, trawls, hand-held rod and reel, and fish-traps; it is rarely caught in offshore pelagic fisheries (Bruce 2008, Lowe *et al.* 2012, Dewar *et al*. 2013, Lyons *et al*. 2013, Francis 2017, Onate-Gonzalez *et al.* 2017). The species has a relatively high post-release survival in net fisheries (Lyons *et al.* 2013, Benson *et al.* 2018). The White Shark is targeted in beach protection programs in Australia and South Africa that use drum-lines and gillnets; however, in some instances these programs release live sharks (Dudley and Simpfendorfer 2006, Bruce 2008, Reid *et al.* 2011, Braccini *et al.* 2017, Kock *et al.* 2018, Lee *et al.* 2018, Roff *et al.* 2018). A shark control program in Réunion Island targets Tiger Sharks (*Galeocerdo cuvier*) and Bull Sharks (*Carcharhinus leucas*), with no captures of White Sharks reported to date (Florida Museum 2019).

Conservation Actions (see Appendix for additional information)

The success of actions agreed through international wildlife and fisheries treaties depends on implementation at the domestic level; for sharks, such follow up actions have to date been seriously lacking. The White Shark was among the first shark species listed under several wildlife treaties. Many fishing nations worldwide and the European Union have domestic regulations specifically aimed at protecting White Sharks. In 2002, the White Shark was listed on Appendix I and II of the Convention on Migratory Species (CMS), which respectively obligates Parties to strictly protect the species and to work regionally toward conservation, specifically through the CMS Memorandum of Understanding for Migratory Sharks. In 2004, the White Shark was added to Appendix II of the Convention on International Trade in Endangered Species (CITES), which requires Parties to ensure that exports be accompanied by permits based on findings that parts are sourced from legal and sustainable fisheries. In 2012, the General Fisheries Commission for the Mediterranean (GFCM) banned retention and mandated careful release for the White Shark and 23 other elasmobranch species listed on the Barcelona Convention Annex II. Implementation by GFCM Parties, however, has been very slow. To prevent overfishing and allow recovery, it is recommended that all White Shark conservation commitments under international wildlife treaties be fully implemented. For CMS Parties, this includes strict protections. In addition, initiatives to prevent lethal contact, minimize bycatch mortality, promote safe release, and improve reporting of catches (including discards) are needed. At a minimum, White Sharks should be subject to catch limits based on scientific advice and/or the precautionary approach.

Credits

Assessor(s): Rigby, C.L., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Herman, K., Jabado, R.W., Jones, G.C.A., Liu, K.M., Lowe, C.G, Marshall, A., Pacoureau, N., Romanov, E., Sherley, R.B. & Winker, H.

Reviewer(s): Bradford, R., Kock, A., Dulvy, N.K. & Kyne, P.M.

Contributor(s): Fergusson, I. & Compagno, L.J.V.

Facilitator(s) and Compiler(s):	Rigby, C.L., Kyne, P.M., Pollom, R., Herman, K. & Dulvy, N.K.
Partner(s) and Institution(s):	Georgia Aquarium
Authority/Authorities:	IUCN SSC Shark Specialist Group (sharks and rays)

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Disclaimer

External Resources

For Supplementary Material, and for Images and External Links to Additional Information, please see the Red List website.

Appendix

Habitats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Habitat	Season	Suitability	Major Importance?
9. Marine Neritic -> 9.1. Marine Neritic - Pelagic	Resident	Suitable	Yes
9. Marine Neritic -> 9.10. Marine Neritic - Estuaries	Resident	Suitable	Yes
10. Marine Oceanic -> 10.1. Marine Oceanic - Epipelagic (0-200m)	Resident	Suitable	Yes
10. Marine Oceanic -> 10.2. Marine Oceanic - Mesopelagic (200-1000m)	Resident	Suitable	Yes
10. Marine Oceanic -> 10.3. Marine Oceanic - Bathypelagic (1000-4000m)	Resident	Suitable	Yes

Threats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Threat	Timing	Scope	Severity	Impact Score
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.1. Intentional use: (subsistence/small scale) [harvest]	Ongoing	Minority (50%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.3. Unintentional effects: (subsistence/small scale) [harvest]	Ongoing	Minority (50%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.4. Unintentional effects: (large scale) [harvest]	Ongoing	Minority (50%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.5. Persecution/control	Ongoing	Minority (50%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		

Conservation Actions in Place

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Action in Place
In-place research and monitoring
Action Recovery Plan: Yes
Systematic monitoring scheme: Yes
In-place land/water protection

Conservation Action in Place
Conservation sites identified: Yes, over part of range
Area based regional management plan: No
Occurs in at least one protected area: Yes
Invasive species control or prevention: Not Applicable
In-place species management
Harvest management plan: Yes
Successfully reintroduced or introduced benignly: No
Subject to ex-situ conservation: No
In-place education
Subject to recent education and awareness programmes: Yes
Included in international legislation: Yes
Subject to any international management / trade controls: Yes

Conservation Actions Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Action Needed
1. Land/water protection -> 1.1. Site/area protection
3. Species management -> 3.1. Species management -> 3.1.1. Harvest management
3. Species management -> 3.1. Species management -> 3.1.2. Trade management
3. Species management -> 3.2. Species recovery

Research Needed

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Research Needed
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
1. Research -> 1.4. Harvest, use & livelihoods
3. Monitoring -> 3.1. Population trends
3. Monitoring -> 3.2. Harvest level trends

Additional Data Fields

Distribution
Lower depth limit (m): 1,200
Upper depth limit (m): 0
Habitats and Ecology
Generation Length (years): 53

Amendment

Amendment reason: This amended assessment corrects the countries of occurrence list (to remove Egypt, Israel and Kuwait, which were incorrectly included in the original assessment) and includes a corrected distribution map (to exclude the Red Sea and the Persian Gulf from this species' range).

The IUCN Red List Partnership

The IUCN Red List of Threatened Species™ is produced and managed by the IUCN Global Species Programme, the IUCN Species Survival Commission (SSC) and The IUCN Red List Partnership.

The IUCN Red List Partners are: ABQ BioPark; Arizona State University; BirdLife International; Botanic Gardens Conservation International; Conservation International; Missouri Botanical Garden; NatureServe; Re:wild; Royal Botanic Gardens, Kew; Sapienza University of Rome; Texas A&M University; and Zoological Society of London.